

Neutral Voice in Echo Chambers: Effects of AI Participation on Group Polarization and Social Perception

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Abstract

Group discussions among like-minded individuals often intensify polarization, producing echo-chamber effects and prompting questions about whether conversational AI can facilitate more rational deliberation. Prior HCI research has predominantly conceptualized AI as a facilitator or support tool, leaving underexplored the social role of AI as an agent for reducing polarization. We report an experimental study examining how a neutral AI agent influences group discussions of a contentious public policy issue in Taiwan—the Digital National ID. Groups of three participants sharing the same initial stance engaged in structured text-based discussions either with or without an AI participant contributing neutral, information-oriented responses. Results indicate that AI participation did not significantly reduce attitude polarization. However, social perception outcomes reveal a notable asymmetry: while the AI was generally perceived as competent, participants holding opposing stances reported significantly higher levels of discomfort toward the AI. These findings suggest that introducing AI as a discussion participant reshapes affective and social dynamics even when attitudinal outcomes remain unchanged. We discuss implications for designing human–AI interaction systems for deliberative contexts

CCS Concepts

• **Human-centered computing**; • **Empirical studies in HCI**; • **Collaborative and social computing**;

Keywords

human–AI interaction, group polarization, deliberation, social perception, AI participation

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1 INTRODUCTION

Generative AI is increasingly moving beyond information retrieval and customer support into public discussion and deliberative settings, where it may participate in multi-turn group interaction as

a member rather than a tool [2, 15, 17]. Despite this shift, most existing systems still position AI as a facilitator or decision aid, and we lack empirical evidence on how AI participation as a discussion member shapes polarization and social dynamics in contentious conversations.

Group polarization is a well-established phenomenon in social psychology, whereby attitudes tend to become more extreme following group discussion, particularly in politically charged or homogeneous settings [6, 8]. Prior work attributes polarization to mechanisms such as persuasive argumentation and social comparison, through which dominant viewpoints gain salience and individuals align their expressed positions with perceived group norms [9, 10]. Importantly, polarization in like-minded groups has proven difficult to mitigate: extreme members can exert disproportionate influence [14], and structured dialogue interventions may yield mixed or counterproductive effects [20].

Recent studies suggest that AI-mediated interventions may improve deliberative interaction under certain conditions by increasing openness to opposing views or reducing hostile language through neutral, evidence-based responses [13, 19, 25]. At the same time, AI participation can provoke aversion or penalty effects, with users reacting more harshly to algorithmic interventions than to comparable human behavior [3], and deliberative contexts producing reduced legitimacy for AI involvement [2]. These findings raise an open question: when AI becomes an embedded discussion member, will it be accepted as a legitimate participant or resisted as a disruptive presence?

To address this gap, we conduct an online group discussion experiment examining the effects of a neutral AI agent embedded as a fourth member in groups. Using a 2×2 between-subjects design manipulating AI presence and initial stance, we investigate both polarization outcomes and participants' social perceptions of the AI using the Robotic Social Attributes Scale (RoSAS) [4] and qualitative thematic analysis [24]. We investigate the following questions:

RQ1: Can AI participation influence group polarization in like-minded group discussions?

RQ2: How do participants perceive the AI's social role and function as a group member?

2 RELATED WORK

2.1 Group Polarization and Deliberative Dynamics

Group polarization describes the tendency for prior attitudes to become more extreme following group discussion, particularly on politically or socially charged issues [8]. Prior research attributes



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polarization to mechanisms such as persuasive argumentation, whereby dominant viewpoints gain salience during discussion [10], and social comparison, whereby individuals adjust their expressed positions to align with perceived group norms [9]. These dynamics are especially pronounced in homogeneous groups, where shared initial stances can amplify confirmation bias and reinforce existing attitudes [6, 8].

Importantly, empirical work suggests that mitigating polarization in real group interactions is difficult. Members with extreme positions often exert disproportionate influence on group outcomes [14], and structured interventions such as dialogue training have shown mixed or even counterproductive effects [17]. Together, these findings suggest that polarization is a socially embedded process that is resistant to short-term or purely informational interventions.

2.2 AI as a Social Participant in Polarized Discussions

As AI systems increasingly participate in collaborative and deliberative settings, research has shown that people tend to apply social expectations and norms to artificial agents, even in the absence of explicit anthropomorphic cues [5]. In group interaction, AI is therefore not perceived solely as a functional tool, but as a social actor whose identity and role shape acceptance and influence [7, 11, 12].

Prior work suggests that AI-mediated interventions can improve aspects of discussion quality under certain conditions, such as increasing openness to opposing views or reducing hostile language through neutral, evidence-based responses [13, 15, 16, 25]. However, AI participation may also provoke resistance or aversion. Users often react more harshly to algorithmic interventions than to comparable human behavior [3], and recent research identifies an “AI penalty” in deliberative contexts, whereby AI involvement reduces perceived legitimacy and acceptance [2]. Moreover, AI systems interacting in polarized or echo-chamber environments may themselves reinforce polarization dynamics [16, 21].

Taken together, existing literature highlights both the potential and the risks of AI participation in polarized discussions. Critically, there remains limited empirical evidence examining AI as an embedded discussion member within like-minded groups, particularly with respect to its social perception and affective consequences. This study addresses this gap by introducing a neutral AI participant to investigate its effects on polarization outcomes and social experience in group deliberation.

3 METHODS

3.1 Experimental Design

We conducted an online group discussion experiment using a 2×2 between-subjects design, manipulating (1) AI participation (AI-present vs. control) and (2) participants’ initial stance toward the discussion topic (pro vs. con). The study focused on deliberation within like-minded groups, a setting known to exacerbate group polarization [6, 8]. Each group consisted of three human participants, with a neutral AI agent introduced as a fourth member in the AI-present condition.

3.2 AI Agent Design

The AI participant was implemented as a retrieval-augmented generation (RAG) system utilizing GPT-4 and was designed to serve as a neutral discussion member rather than a moderator or authority figure. The AI generated informational, evidence-based responses while refraining from expressing evaluative judgments or explicit stances. In the AI-present condition, the agent contributed once per discussion round following the contributions of all human participants. In the control condition, discussions proceeded without AI participation, and only human participants engaged in the conversation.

The following excerpt illustrates a typical AI response during discussion:

Human participant: “Digital ID could improve efficiency but raises privacy risks.”

AI agent: “Digital identity systems have been associated with improved administrative efficiency in several national implementations. However, privacy concerns, including risks related to data breaches and surveillance, have also been widely discussed. Policymakers often consider both efficiency benefits and privacy safeguards when evaluating such systems.”

This example reflects the AI’s neutral, informational, and non-persuasive conversational style.

Pilot Study. Prior to the main experiment, we conducted a pilot study with 20 participants to evaluate the perceived neutrality, naturalness, and conversational appropriateness of the AI agent’s responses. Participants were presented with anonymized excerpts of group discussions containing both human and AI responses and rated each speaker on neutrality, logical coherence, naturalness, and conversational responsiveness using Likert scales. A one-way ANOVA revealed no significant differences between AI and human speakers across evaluation dimensions ($p > .05$), indicating that the AI agent’s conversational behavior was perceived as comparable to that of human participants.

3.3 Participants

A total of 80 participants completed the study and were assigned to one of the four experimental conditions defined by AI presence and initial stance. All participants completed the full discussion procedure as well as the pre- and post-discussion questionnaires.

3.4 Procedure

Participants first completed a pre-discussion questionnaire to assess their existing attitudes toward the implementation of a “Digital National ID.” This topic represents a contentious issue, as it involves the trade-off between digital surveillance and civil privacy. Citizens in Taiwan are divided into two camps: opponents advocate for strengthening protections for personal privacy, freedom, and autonomy, arguing that digital identity systems may lead to government overreach and information misuse, thereby undermining democratic values and freedom of speech. Conversely, proponents contend that implementing a digital identity system would enhance efficiency, strengthen governance capacity, and prevent crime amid increasing cybersecurity threats and public safety challenges.

Following completion of the questionnaire, participants were then assigned to a discussion group and engaged in a structured

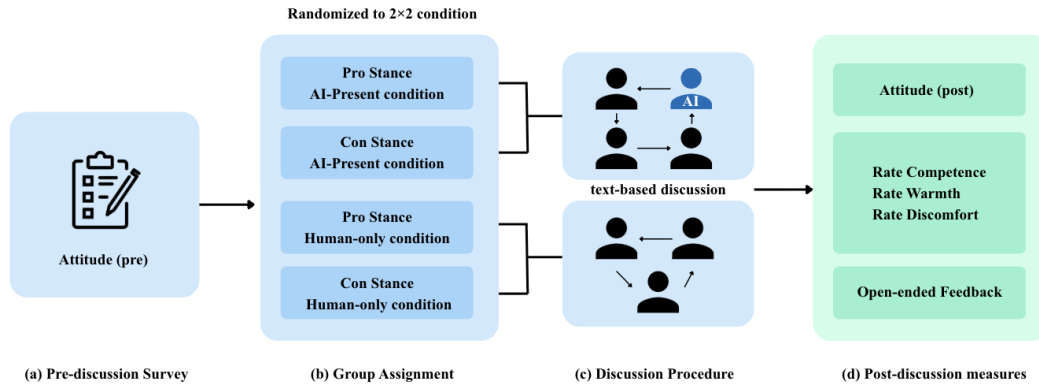


Figure 1: Experimental Procedure. (a) Pre-discussion survey: Participants completed a pre-test assessing their stance on the policy issue. (b) Group Assignment: Based on stance (Pro/Con), participants were randomly assigned to one of two discussion conditions (AI-present condition vs. Human-only condition), resulting in a 2×2 between-subjects design. (c) Discussion Procedure: Groups of three human participants discussed the topic synchronously for 30 minutes, with a AI agent included in the AI-present condition. (d) Post-discussion measures: Participants completed follow-up measures including attitude reassessment, the ROSAS, and open-ended feedback. (Fig. 1)

text-based discussion consisting of seven rounds via an online platform. In each round, all three human participants contributed sequentially, followed by a single response from the AI agent in the AI-present condition. In the control condition, discussions proceeded without AI intervention. Upon conclusion of the discussion, participants completed a post-discussion questionnaire measuring attitude change and social perceptions. (Fig. 1)

This study was conducted as part of a university course project and adhered to standard ethical practices. All participants were informed of the study’s purpose, participated voluntarily. No personally identifiable information was stored.

3.5 Measures and Analysis

Group polarization was assessed using a Likert-scale attitude measure administered before and after discussion, operationalized as changes in attitude extremity [1, 6, 8, 18]. Social perception of the AI agent was measured using the Robotic Social Attributes Scale (RoSAS), capturing competence, warmth, and discomfort [4]. Discussion transcripts and open-ended responses were analyzed qualitatively using thematic analysis [24].

4 RESULTS

4.1 Effects of AI Participation on Group Polarization

We examined whether the presence of an AI discussion member influenced changes in attitude extremity from pre- to post-discussion. Descriptive results showed that participants’ attitudes remained largely stable across conditions. A repeated-measures ANOVA revealed no significant main effect of time (pre- vs. post-discussion) ($F(1,25)=0.009, p=0.924$) and no significant main effect of AI participation ($F(1,25)=0.009, p=0.926$). (Fig. 2)

Together, these results indicate that introducing a neutral AI participant did not produce a statistically significant change in

attitude extremity. These findings suggest that, within the scope of this study, we did not find evidence that AI participation influenced polarization outcomes in short-term discussions.

4.2 Social Perception of the AI Agent Effects of AI Participation on Group Polarization

Participants’ perceptions of the AI agent were assessed using the Robotic Social Attributes Scale (RoSAS). Overall, the AI was perceived as relatively high in competence, suggesting that participants generally regarded the agent as capable and informative. In contrast, ratings on the discomfort dimension exhibited notable variation across conditions ($t(35)= 2.73, p = .021$). (Fig. 3)

Specifically, participants holding opposing initial stances reported **significantly higher levels of discomfort toward the AI agent** compared to those whose initial stance aligned with the group discussion. This pattern indicates that while the AI’s informational role was broadly acknowledged, affective responses to its presence were shaped by participants’ prior positions.

4.3 Qualitative Insights into AI Participation

Thematic analysis of participants’ open-ended responses and discussion transcripts revealed nuanced interpretations of the AI agent’s role. Although the AI was designed to provide factual clarifications and neutral summaries, its social positioning within the group was frequently questioned.

Participants whose initial stance aligned with the majority (in-group) often perceived the AI’s input as supportive, describing it as offering “informative additions” or “helpful background” that reinforced group coherence. In contrast, participants with opposing stances (outgroup) reported greater discomfort, describing the AI as “out of sync” or “socially detached” from the conversational flow. These participants often characterized the AI as an external

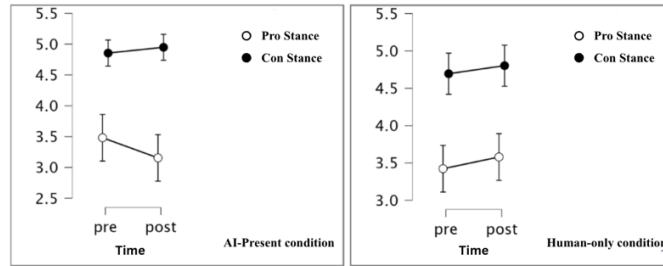


Figure 2: Changes in attitude extremity from pre- to post-discussion across conditions. The left panel shows the AI-present condition, and the right panel shows the human-only condition. White circles indicate participants with a pro stance, and black circles indicate participants with a con stance. (Fig. 2)

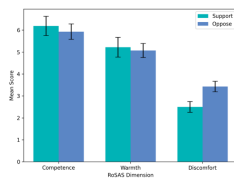


Figure 3: Participants' ratings of the AI on competence, warmth, and discomfort. Notably, discomfort varied by stance alignment. (Fig. 3)

presence rather than a group member, perceiving its contributions as intrusive or misaligned with emergent group norms.

Crucially, the discomfort expressed by outgroup members did not stem from the AI's content accuracy, but from its ambiguous social role. Some participants speculated about the AI's intentions or affiliation, suggesting that its perceived neutrality failed to establish relational trust. This stance-dependent interpretation of the AI highlights the importance of affective and relational factors in shaping user experiences with artificial agents in group deliberation.

5 DISCUSSION

5.1 Boundary Conditions of AI-Driven Polarization Mitigation

This study explored whether a neutral AI agent could reduce attitudinal extremity in like-minded group discussions. Despite expectations that AI mediation might temper polarization, our results showed no significant effect on post-discussion attitude shifts. This aligns with prior findings that group polarization is deeply rooted in shared identity and normative reinforcement, making it resistant to short-term informational interventions [6, 8].

Rather than signaling ineffectiveness, the null result highlights important boundary conditions. In ideologically aligned groups, the informational contributions of an AI—no matter how accurate or balanced—may lack sufficient social capital to influence group trajectories. This suggests that mitigating polarization may require

more than just neutral input; it may demand social legitimacy and relational embeddedness.

5.2 Role Ambiguity and Affective Friction

Although the AI agent had limited influence on attitudinal outcomes, it significantly shaped participants' social and emotional responses. Notably, those whose stance diverged from the group expressed elevated discomfort toward the AI. Qualitative analysis suggests this discomfort arose less from message content and more from role ambiguity—participants were unsure whether the AI functioned as a peer, a facilitator, or an external monitor.

This lack of role clarity disrupted group norms, especially for those already feeling outnumbered. Without clear cues about the AI's alignment or function, participants projected their social uncertainty onto the system. These dynamics reveal how human-AI interaction is not only shaped by content, but by how technology is socially situated.

5.3 Design Implications for Group-Embedded AI

Designing AI systems for group deliberation demands more than conversational fluency. Our findings suggest that social legibility—users' ability to infer the AI's role and intent—is critical for emotional acceptance. Prior work has emphasized transparency and consistency in AI behavior [23], but our study extends this by showing that role signaling is especially crucial in group contexts.

To avoid affective friction, designers should make the AI's role explicit (e.g., "observer," "informational aid") and match language style and timing to group norms. Future systems could adaptively signal alignment or disengagement based on group dynamics, helping users contextualize the AI's function without disrupting social cohesion.

5.4 Limitation and Future Work

This controlled study used brief online discussions with a fixed-response AI. Future work should test longer-term, naturalistic settings (e.g., asynchronous or cross-stance groups) and explore adaptive AI roles, such as personalized empathy or dynamic tone shifts. Future systems could explore adaptive role framing, in which AI dynamically adjusts its communicative style or explicitly signals its function (e.g., informational assistant vs. discussion observer) based

on group dynamics and participant responses to reduce ambiguity and improve group cohesion.

6 CONCLUSION

This study examined the effects of a neutral AI participant on group polarization and social perception during online discussions. While AI participation did not significantly alter attitude extremity, it had a meaningful impact on how participants experienced the discussion. In particular, discomfort toward the AI varied systematically with participants' stance alignment, underscoring the importance of perceived group membership and social positioning in human–AI interaction.

Our findings emphasize that the presence of AI in group settings carries social and emotional consequences, even when cognitive outcomes remain unchanged. Designing AI for deliberation thus requires attention not only to information quality but to social role clarity and affective reception. By identifying role ambiguity as a key tension point, this work contributes to a more relational understanding of AI integration in human groups.

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